Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B.Tech.

MAKEUP SEMESTER EXAMINATION, DECEMBER – 2024

ELE 3127 - Foundations EV and Hybrid Vehicles

Time: 3 Hours **Date: DD/MM/YYYY** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A.	Examine the policies and incentives that have driven the growth of the Electric Vehicle (EV) market in India.	***4***
1B.	Which Indian states are leading in EV adoption, and what initiatives have they implemented?	***2***
1C.	Compare the market strategies of established EV manufacturers with new entrants in India	***4***
2A.	Explain how insulation design impacts the safety of high-voltage systems in Electric Vehicles (EVs).	***3***
2B.	Discuss the role of battery management systems (BMS) in ensuring the safety of high-voltage EV components.	***2***
2C.	How does thermal runaway in EV batteries affect safety, and what preventive measures are used to mitigate it?	***5***
3A.	Compare the energy efficiency of Hybrid Electric Vehicles (HEVs) with that of conventional Internal Combustion Engine (ICE) vehicles.	***4***
3B.	What technological advancements distinguish modern PHEVs from earlier generations of HEVs?	***3***
3C.	How do regenerative braking systems enhance the performance of HEVs and PHEVs?	***3***
4A.	If a 15Ah battery is discharged at a 0.8C rate, calculate the discharge current and determine how long the battery will last.	***2***
4B.	Discuss the implications of mining raw materials for EV batteries on local ecosystems and global sustainability.	***4***
4C.	Examine the potential of recycling technologies in reducing the environmental footprint of EV batteries.	***4***
5A.	How do Permanent Magnet Synchronous Motors (PMSMs) contribute to improving EV efficiency at high speeds?	***4***
5B.	Compare the noise, vibration, and harshness (NVH) characteristics of Brushless DC (BLDC) motors and PMSMs in electric vehicles.	***4***
5C.	What are the advantages and limitations of using Induction Motors versus PMSMs in EV applications?	***2***